

Liuting Chen

jnk758@alumni.ku.dk | +45 7888 3143

RESEARCH INTEREST

Causal Inference, Representation Learning, Multi-Agent Communication, Human-AI Interaction.

EDUCATION

M.Sc. in Mathematics, University of Copenhagen

Grade: 8.71/12

Sep. 2025 – Jan. 2027 (Expected)

Copenhagen, Denmark

B.Sc. in Mathematics, Southwest Jiaotong University

GPA: 3.3/4

Sep. 2021 – Jun. 2025

Chengdu, China

PUBLICATIONS AND MANUSCRIPTS

L. Chen and R. Xu, "A Dynamic-Static Dataset of Facial Feature Variations Induced by Six Basic Tastes," the IEEE 20th International Conference on Automatic Face and Gesture Recognition (FG), Kyoto, Japan, 2026, pp. 1-10, doi: 10.1109/FG67764.2026.11556943.

L. Chen and M. Alex, "I-FLOP: Fast Learning of Order and Parents from Interventional Data," **Oral** presentation at the 13th International Conference on Probabilistic Graphical Models (PGM), Valencia, Spain, 2026.

RESEARCH EXPERIENCE

Master's Thesis: Formal Agent Benchmarking and Statistical Reliability for GPAP-Tools Evaluation

Supervisors: [Harrison Bohua Zhu](#) (Prof., UCPH), [Jun Yang](#) (Prof., UCPH)

Aug. 2026 – Dec. 2026 (Expected)

- Developing an execution-aware framework for analyzing tool-using AI agents through multi-step interaction trajectories, including structured actions, tool interactions, intermediate states, and task outcomes.
- Developing a fine-grained diagnostic framework for sequential agent behavior, distinguishing failures in reasoning and action selection from those arising through retrieval and tool interaction.
- Designing small-sample experiments and intervention-based analyses to identify failure mechanisms and compare behavioral trajectories across AI agents.

Summer Research: Source-Information Leakage in LLM-as-a-Judge Systems

Visiting Researcher, [Global Pathogen Analysis Platform \(GPAP\)](#)

Supervisor: [Harrison Bohua Zhu](#) (Prof., UCPH)

Jul. 2026 – Oct. 2026 (Expected)

- Developed a dynamic causal framework for source-information leakage in LLM-as-a-Judge systems, leading to a first-author manuscript in preparation for submission to ICLR 2027.
- Formalized source-information leakage as a causal process, distinguishing source availability, observed exposure, causal utilization, and verified mitigation.
- Designed intervention-based auditing experiments and derived finite-sample statistical guarantees for identifying and quantifying causal information leakage.

Research Project: I-FLOP: Fast Learning of Order and Parents from Interventional Data
Supervisor: [Alex Markham](#) (*PostDoc., [Copenhagen Causality Lab](#)*)
Dec. 2025 – May 2026

- Developed a new algorithm for learning Gaussian Bayesian networks from observations and hard interventions, leading to a first-author accepted oral presentation at *PGM 2026*.
- Developed a score-based structure-learning approach that exploits invariance across intervention environments, with theoretical guarantees for local parent recovery and global I-MEC consistency.
- Compared with GIES and I-GSP, I-FLOP achieved lower I-CPDAG structural Hamming distance (SHD) and a favorable accuracy-runtime trade-off.

Bachelor's Thesis: A Dynamic-Static Dataset of Facial Feature Variations Induced by Six Basic Tastes
Supervisor: Renyu Xu (*Lect., SWJTU*)
Dec. 2024 – Jun. 2025

- Designed TasteFace, a human-behavior dataset capturing spontaneous and temporally evolving facial responses to controlled sensory stimuli, resulting in a first-author publication at IEEE FG 2026.
- Designed and conducted controlled experiments with 48 participants, collecting dynamic facial-response recordings under standardized stimulus conditions.
- Developed computer-vision models to study representation learning from dynamic human behavioral data under limited-data conditions.

Statistical Modeling Competition: Carbon-Emission Forecasting
Mar. 2024 – May 2024

- Received the Third Prize in the 10th National College Student Statistical Modeling Competition.

APPLIED DATA EXPERIENCE

Noldus Information Technology Co., Ltd
Jul. 2024 – Sep. 2024

- Processed and integrated multimodal human-behavior data across questionnaire responses, EEG, facial expressions, images, and video-derived features from controlled experimental studies.
- Maintained and evaluated sequence-modeling pipelines, including LSTMs for learning temporal patterns from multimodal behavioral data.
- Standardized the organization of datasets, training scripts, test configurations, model outputs, and computing environments to support reproducible analysis and reliable project handover.

TECHNICAL SKILLS

Programming: Python, R, MATLAB
Languages: Mandarin Chinese (native), English (IELTS Academic: 6.5)